



A Sensitive and Self-Regenerable Particulate Matter Sensor with a H_3PO_4 -Modified $\text{BaZr}_{0.8}\text{Y}_{0.2}\text{O}_{3-\delta}$ Electrolyte and an IrO_2 -Catalyzed Sensing Electrode

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There have been many attempts to develop solid-state sensor devices for detecting particulate matter (PM) in diesel exhaust; however, in most of these, the accumulated PM must be burned intermittently to allow subsequent sensing cycles. Here, we report a self-regenerable PM sensor using a proton conductive solid electrolyte and an active working electrode for PM oxidation. The withstanding voltage capability of $\text{BaZr}_{0.8}\text{Y}_{0.2}\text{O}_{3-\delta}$ was greatly improved by the growth of a dense $\text{Zr}_{1-x}\text{Y}_x\text{P}_2\text{O}_7$ film on the electrolyte surface. The reaction of PM with active oxygen under anodic polarization was further enhanced by the addition of IrO_2 to the working electrode. As a result of these combined modifications, when the working electrode was anodically polarized, PM was oxidized to CO_2 according to a four-electron reaction ($\text{C} + 2\text{H}_2\text{O} \rightarrow \text{CO}_2 + 4\text{H}^+ + 4\text{e}^-$) while remaining below the self-ignition temperature. This amperometric sensor successfully produced a current signal corresponding to the quantity of PM in a gas stream at an operating temperature of 300°C . These results demonstrate that the sensor can carry out continuous monitoring of PM concentrations while self-regenerating.

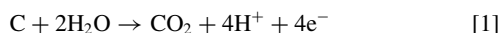
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Hazards to human health due to the particulate matter (PM) emitted from diesel engines has become a subject of great concern.¹ The ability to measure PM from diesel exhausts is useful for the reduction of PM emissions through combustion control in engines and on-board diagnostics for PM filters.² Conventional technologies that can be used for PM monitoring in exhaust flows include electrical insulator- and semiconductor-based sensors.^{3–6} Both types of sensor apply a high voltage between two electrodes and measure the current signal that is correlated with a specific PM concentration. However, these sensors are subject to baseline drift due to the accumulation of PM on the electrodes or in the gap between them. A conventional solution to this problem is to burn off the accumulated PM by increasing the temperature of the sensor.

An alternative approach is to utilize electrochemical oxidation of carbon by active oxygen for PM sensing.^{7–9} An electrochemical cell consisting of a proton-conducting $\text{Sn}_{0.9}\text{In}_{0.1}\text{P}_2\text{O}_7$ electrolyte and a catalytically active Pt working electrode produced active oxygen at the working electrode during a water vapor oxidation reaction. This oxygen species exhibited high activity for carbon oxidation at temperatures of 50°C or higher. Interestingly, the formation rate of CO_2 increased with increasing temperature, reaching the theoretical value calculated from the following four-electron reaction at 200°C .



Such a high faradaic efficiency successfully produced a current signal corresponding to the quantity of PM in a bench test apparatus.⁹ In addition, Reaction 1 self-regenerated the working electrode, which enabled the PM concentration to be monitored in real time to a significant extent. However, there is a concern that the Sn^{4+} ions of the electrolyte are electrochemically reduced to metallic Sn ($E_0 = -0.13$ V vs. SHE) at the counter electrode side under cathodic polarization, which may ultimately form continuous charge pathways between the two electrodes. An additional concern is the oxidation of metallic Pt in the working electrode to high-valent ions ($E_0 = 0.98, 1.04, \text{ and } 1.70$ V vs. SHE for Pt^{2+} , Pt^{4+} , and Pt^{6+} , respectively) under anodic polarization, significantly reducing the catalytic activity of Reaction 1. Thus, the synthesis of both electrolyte and electrode catalyst with high withstanding voltage is a crucial requirement for practical application of the proposed sensor.

In this study, two approaches to meeting the above requirement were investigated. First, we grew a thin, dense $\text{Zr}_{1-x}\text{Y}_x\text{P}_2\text{O}_7$ film on the surface of a $\text{BaZr}_{0.8}\text{Y}_{0.2}\text{O}_{3-\delta}$ ceramic substrate, which provides a good balance between withstanding voltage and proton conductivity. We next added IrO_2 or RuO_2 , both of which are known electrocatalysts for water oxidation,^{10–13} with a Pt working electrode. Since these metal oxides also have excellent proton exchange ability,^{14–16} a three-dimensional network of protons was expected to form within the working electrode. The goals of the present work are: (1) to suppress both electron and hole conduction in the electrolyte under polarization, (2) to compare various electrocatalysts in terms of attainable sensitivity toward PM, and (3) to examine the validity of these approaches as a means of improving the PM sensor operating at an applied voltage of 3 V and in the temperature range of $240\text{--}350^\circ\text{C}$.

Experimental

Materials.—A $\text{Zr}_{1-x}\text{Y}_x\text{P}_2\text{O}_7$ film was synthesized on the surface of a sintered pellet of $\text{BaZr}_{0.8}\text{Y}_{0.2}\text{O}_{3-\delta}$ as a substrate (diameter: 13 mm; thickness: 1 mm) as follows. Initially, 4 mol% ZnO powder was mixed with $\text{BaZr}_{0.8}\text{Y}_{0.2}\text{O}_{3-\delta}$ powder (Kojundo Chemical Laboratory) as a sintering aid.¹⁷ The mixture was ground in ethanol using a planetary ball mill at 200 rpm for 1 h. After drying in an oven at 120°C for 10 h, the powder was uniaxially pressed into pellets under a pressure of 100 MPa for 1 min and then sintered at 1350°C for 10 h in air (this material is hereafter denoted BZY_ZnO). The thickness of the resulting substrate pellets was adjusted to 1 mm by polishing the surface with abrasive paper. The substrate was immersed in 85% H_3PO_4 solution and then heated to 500°C at 4°C min^{-1} , held at this temperature for 5 minutes, and cooled to room temperature at 4°C min^{-1} . Finally, the product was thoroughly washed in deionized water using an ultrasonic cleaner for 20 minutes and dried at 100°C for 2 h (this material is hereafter denoted BZY_ZnO_3HPO₄). $\text{BaZr}_{0.8}\text{Y}_{0.2}\text{O}_{3-\delta}$ electrolyte sintered at 1650°C without the use of ZnO (hereafter denoted BZY) and a $\text{Zr}_{0.84}\text{Y}_{0.16}\text{P}_2\text{O}_7$ pellet electrolyte isostatically pressed at 100 MPa (hereafter denoted ZYPO) were also used as electrolytes for comparison studies.

IrO_2 (Wako Chemicals), RuO_2 , (Aldrich), $\text{BaZr}_{0.8}\text{Y}_{0.2}\text{O}_{3-\delta}$, $\text{Sn}_{0.9}\text{In}_{0.1}\text{P}_2\text{O}_7$, and WP_2O_7 powders were used as both electrocatalysts and ionomers for the working electrode. The preparation of the two latter compounds has been described in detail elsewhere.^{18,19} A

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